Thanh Tai Duong (published as Thanh-Tai Duong)

Ph.D. | Therapeutic Medical Physicist | Researcher | Lecturer

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Google Scholar, LinkedIn, and ResearchGate profiles are linked from the official website.

A concise academic profile for prospective collaborators, faculty, research teams, clinical employers, and scientific editors. The full CV and complete publication record are available on the official website.

Current profile

Clinical role	Medical Physicist, Jaeger Corporation / Spencer Hospital, United States (2025-present)
Research role	Research Associate, Robarts Research Institute, Western University, Canada (2024-present)
Academic role	Lecturer / Academic Collaborator, Nguyen Tat Thanh University, Vietnam (2022-present)

Publication name variants: Duong Thanh Tai and Thanh Tai Duong.

Research themes

Radiation therapy physics

Treatment planning, IMRT and VMAT, SBRT and SRS support, brachytherapy physics, commissioning, and quality assurance.

Monte Carlo dosimetry

Physics-based dose calculation, verification, and simulation.

Medical imaging

CBCT and CT workflows, calibration, image processing, deep learning, and quantitative imaging.

Radiotheranostics

Patient-specific dosimetry and modeling for targeted radionuclide therapy and combined treatment strategies.

Selected expertise

- External-beam radiotherapy and treatment-plan review
- Patient-specific QA, machine QA, and imaging QA
- HDR/LDR brachytherapy and Halcyon commissioning
- EGSnrc and TOPAS Monte Carlo simulation
- Personalized radiopharmaceutical dosimetry
- Deep learning, radiomics, and medical physics education

Academic and professional record

Education, appointments, service, and collaboration

Education

- Ph.D. in Medical Physics, VNUHCM-University of Science, Vietnam (2015-2019)
- M.Sc. in Nuclear Physics, VNUHCM-University of Science, Vietnam (2010-2012)
- B.Sc. in Applied Physics, VNUHCM-University of Science, Vietnam (2005-2009)

Previous appointments

- Postdoctoral Associate, Robarts Research Institute, Western University, Canada (2022-2024)
- Lecturer, HCMC University of Technology and Education, Vietnam (2020-2022)
- Clinical Medical Physicist, Dong Nai General Hospital, Vietnam (2009-2020)

Editorial and professional service

- Associate Editor, Medical Physics (AAPM/Wiley) (2025-2028)
- Associate Editor, Radiation Physics and Chemistry (Elsevier) (2025-present)
- Editorial Board Member, Biomedical Physics & Engineering Express (IOP Publishing)
- Reviewer for scientific journals in medical physics, radiation oncology, imaging, and radiation science
- Member of AAPM, COMP, VSMP, IRPS, VPS, and EFOMP

Teaching and mentorship

Teaching and academic collaboration have included biomedical imaging technology, image processing, biomedical information systems, physics laboratory, biophysics, and capstone projects. The current CV records supervision and mentorship of more than 25 undergraduate thesis and research projects.

Research collaborations

- Deep learning and explainable-AI classification of mammographic lesions for early breast cancer detection, Institut Pertanian Bogor, Indonesia (2026)
- Monte Carlo-based investigation of hybrid HDR brachytherapy and VMAT for tongue cancer, Universiti Sains Malaysia

Links

Official website: https://thanhtaiphys.github.io/
Full CV: download current CV
Publications: complete publication records